Chinese Journal of Aeronautics, (2025), 38(3): 102982

Chinese Society of Aeronautics and Astronautics
& Beihang University

Chinese Journal of Aeronautics

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REVIEW

Swarm intelligence: A survey of model classification and applications

Chao WANG [a], Shuyuan ZHANG [a], Tianhang MA [a], Yuetong XIAO [a], Michael Zhiqiang CHEN [b], Lei WANG [a,*]

[a] School of Automation Science and Electrical Engineering, Beihang University, Beijing 100191, China
[b] School of Automation, Nanjing University of Science and Technology, Nanjing 210094, China

Received 20 November 2023; revised 28 December 2023; accepted 28 February 2024
Available online 20 March 2024

KEYWORDS

Swarm intelligence;
Ant colony optimization;
Self-driven particle models;
Pheromone communication models;
Leadership decision models;
Empirical research models

Abstract Swarm Intelligence (SI) is a collective behavior that emerges from interaction between individuals in a group. Typical SI includes fish schooling, ant foraging, bird migration, and so on. A great deal of models have been introduced to characterize the mechanism of SI. This article reviews several typical models and classifies them into four categories: self-driven particle models, with Boids model as the primary example; pheromone communication models, including the ant colony pheromone model which serves as the foundation for ant colony optimization; leadership decision models, utilizing the hierarchical dynamics model of pigeon flock as a prime instance; empirical research models, which employ the topological rule model of starling flock as a classic model. On this basis, each type of model is elaborated upon in terms of its typical model overview, applications, and model evaluation. More specifically, multi-agent swarm control, path optimization and obstacle avoidance, formation and consensus control, trajectory tracking in the dense crowd and social networks analysis are surveyed in the application of each category, respectively. Furthermore, the more precise and effective modeling techniques for leadership decision and empirical research models are described. Limitations and potential directions for further exploration in the study of SI are presented.

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1. Introduction

SI has its origin in the observation and study of social behavior in gregarious animals. Its distribution, flexibility, and robustness advantages have provided new solutions to many challenging complex problems. Initially proposed by Beni and Wang to describe the dynamics of swarm robots, SI has, since evolved, been defined in various ways across different fields. [1]

* Corresponding author.
E-mail address: lwang@buaa.edu.cn (L. WANG).
Peer review under responsibility of Editorial Committee of CJA.

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https://doi.org/10.1016/j.cja.2024.03.019
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In natural science, it is typically defined as a collective behavior that emerges from interaction between individuals in a group, whether they are animals or cells. SI refers to the use of algorithms and techniques that involve multiple intelligent agents collaborating, such as robots or sensor nodes.[2] In social science, it is defined as the intelligent behavior exhibited by groups of humans through their interactions and collaborations.[3]

This article primarily focuses on SI within the natural sciences, where the intelligence of a group often surpasses that of its members. This phenomenon is evident in various biological clusters,[4,5] such as fish schooling,[6–8] ant foraging,[9–11] wolf predation,[12] and bird migration.[13] Despite individual limitations, these groups demonstrate orderly self-organization and coordination behaviors at the group level, displaying traits such as distribution, self-organization, cooperation, stability, and adaptability.[14]

Researchers have extensively studied the unique wisdom of biological clusters by establishing models, to uncover the internal mechanisms of SI through simulating the collective behaviors. One of the earliest models, the Boids model by Reynolds,[15] summarized three fundamental rules governing collective behavior. Since then, research has been focused on simulating SI by designing group interaction rules over a long period. For instance, typical models such as swarm model[16] and R-A model[17] were both developed based on Boids model, and the well-known Vicsek model[18] was established from a physical perspective. In addition to the models established by the rules, there are some models based on pheromones, such as the ant colony models[19–21] and bees models.[22] However, most of these models were developed in the absence of empirical data due to the limitations in data collection technology over the past decades. Until the development of tracking positioning and sonar detection technology, this issue has been resolved. This has made it more convenient to observe the behavior of biological clusters and gain a deeper understanding of the spatial aggregation and order of movement of individuals in clusters,[14] for example, the model of starling flocks[23] and the model of fish shoals[24]. Recently, advancements in machine learning have led to increased interest and progress in data-driven models that utilize observed data to yield the qualitatively same collective behavior as exhibited by real swarm.[25–27] This integration of technology and biological insights marks a promising direction for further understanding the wisdom of biological clusters and their internal mechanisms. Since data-driven models are constructed based on real-world data, they can be regarded as a specific category of empirical research models.

Building upon the insights gained from these biological cluster behavior models, corresponding algorithms have been formalized and are referred to as biology-based algorithms[28] in various literature, or simply swarm intelligence algorithms.[29] They can be categorized into six groups based on the number of algorithms used and their biological inspiration: P-article Swarm Optimization (PSO),[30] Artificial Fish Swarm (AFS),[31] Ant Colony Optimization (ACO),[32] Artificial Bee Colony (ABC),[33] Bacterial Foraging Optimization (BFO),[34] and other swarm intelligence algorithms.[35,36] These heuristic optimization algorithms have gained widespread adoption and are employed in diverse applications, including parameter optimization,[37] system identification,[38] robots[39] and other optimization problems of engineering fields.[40] While our focus is primarily on reviewing swarm intelligence models rather than the algorithm, we will discuss some of them in subsequent content.

In addition to algorithms inspired by biology, many applications are directly derived from animal collective behaviors. Unlike the aforementioned algorithm applications in optimization problems, these applications primarily refer to cooperative unmanned cluster systems, including aerial, ground, and marine robotics swarms. In aerial robotics swarm, for example, the task assignment algorithm and interaction mode should be carefully designed to support the cooperative decision-making process of Unmanned Aerial Vehicle (UAV) swarm, which is recently inspired by biological swarm behaviors in various applications.[41] Besides, UAV swarms take actions to carry out assigned tasks through various configurations of formation control. There are several key technologies necessary for autonomous flight, including formation maintenance,[42,43] formation reconfiguration,[44,45] and obstacle avoidance.[46] In ground robotics swarm, for example, with the enhanced development of Unmanned Ground Vehicle (UGV) systems, the exploration and verification of swarm cooperative algorithms have been further deepened. Particle robotics are controlled to achieve autonomous locomotion, phototaxis, and object transportation through a simple distributed algorithm inspired by collective cell migration phenomena in biology.[47] In the field of marine robotics swarm, researchers have conducted extensive research to investigate communication technology,[48] navigation, and location technology[49] inspired by collective behaviors in biology. Particularly, it should be noted that using the swarm algorithms to control robotics is different from simulating a swarm. The former aims to achieve a specific task or goal, while the latter aims to understand the mechanisms of collective behavior and interaction. Hence, studying the swarm intelligence models, algorithms and applications holds immense academic and practical value. The hierarchical distribution of swarm intelligence is shown in Fig. 1.[15–21,30–36,41–49]

Several literatures[50–52] have reviewed SI models, algorithms, and applications because of their excellent performance in various problems. They often reviewed SI models from a specific perspective, such as a physical perspective[53] or a biological perspective.[54] The reviews about SI algorithms and applications usually focus on representative algorithms, such as PSO and ACO, and specific application problems, such as Internet of Things (IoT) and features seletion.[55–58] However, there is seldom a timeline or appropriate classification in the literature reviews regarding models, and there is limited exploration and summarization of the underlying models in the reviews of algorithms and applications. In other words, there is rare research that has reviewed SI algorithms and applications based on underlying models, which is the gap that this article aims to fill. The main contributions of this article are shown as follows:

(1) This article provides a new categorization of typical swarm intelligence models. We organized typical models based on their chronological emergence, starting from time when the most representative Boids model appeared up to the present. Following this, we found that the emergence of representative models during different periods allowed us to divide our focus timeline into four distinct periods. We summarized the common

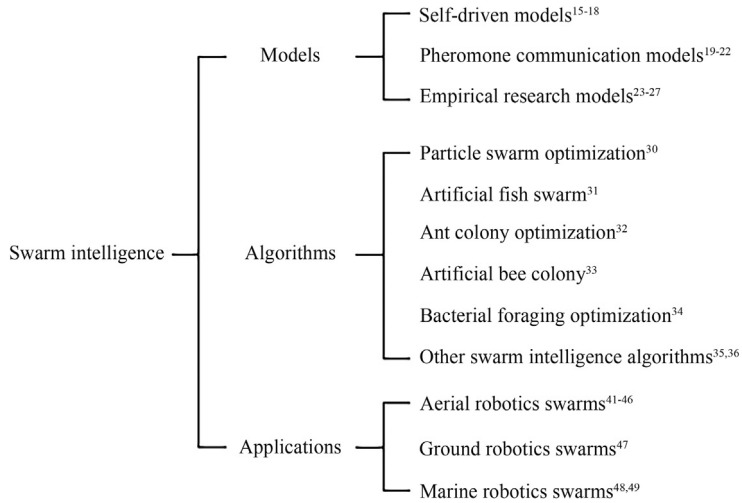

Fig. 1 Hierarchical distribution of swarm intelligence.

characteristics of representative models within each period and classified the existing typical models into four categories: self-driven particle models, pheromone communication models, leadership decision models and empirical research models.

(2) Each type of model is elaborated upon its typical model overview and applications. In section of self-driven particle models, we overview typical models and their applications in multi-agent swarm control. Then, we delve into PSO, summarize its improvement methods and overview its applications in the aerospace area. In section of pheromone communication models, we focus on representative ant colony models, summarize ACO and its improved version, and explore its applications in path optimization and obstacle avoidance. In section of leadership decision models, we introduce typical models and outline methodologies for leadership inference, emphasizing their application in formation and consensus control. In section of empirical research models, we elaborate on small-scale and large-scale cluster models. Moreover, we summarize techniques for constructing empirical research models and survey their applications in trajectory tracking within dense crowds, as well as social network analysis.

The rest of this article is structured as follows: in Section 2, the concept of SI is introduced, and the existing typical models of SI are divided into four categories, including self-driven particle models, pheromone communication models, leadership decision models and empirical research models. Each type of model is elaborated upon in terms of its typical model overview, related applications, and model evaluation from Section 3 to Section 6, respectively. The conclusion and future research direction are presented in Section 7.

2. Classification of typical models of SI

Many social animals, such as ants, birds, and bees, can carry out complex tasks through the interaction and coordination of many individuals, without the need for a centralized control mechanism. Fig. 2 shows the SI of ants, birds and bees. The study of SI aims to identify the mechanisms underlying collective behaviors and to develop algorithms and techniques that can be used to design intelligent, adaptive, and robust systems. One of the most important tasks is to establish models to identify the mechanisms. A variety of models have been developed for SI. It first aroused the interest of researchers in the field of computer graphics,[15] and subsequently became the subject of a great deal of physics research and applied mathematics.[59,60] The goal of establishing SI models is to clarify how individuals form group-level collective behavior through local interaction.[61] Because groups involve a large number of discrete individuals, many physicists consider it in terms of statistical mechanics. This methodology treats SI as an emergent property, resulting from the interactions of numerous individuals conceptualized as particles. The primary focus is on constructing microscopic models to discern the determinants of group-level dynamics. This bottom-up approach, emphasizing the self-organizing nature of groups, allows for a stable model of distributed control, where interactions are limited yet collectively shape coherent group behaviors. This framework not only facilitates nuanced exploration of SI but also aligns with principles of self-organization, making it a robust model for studying complex systems.

An alternative approach is top-down modeling, where the focus lies in initiating the modeling process with the generation of group-level behaviors, proposing simplified models to repli-

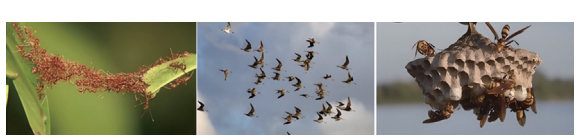

Fig. 2 SI of ants, birds and bees (https://lib.ia.ac.cn/news/newsdetail/68504).

cate this behavior. Top-down models aim to elucidate the workings of a clustered system rather than delving into the underlying reasons for its functionality. While theoretical models inherently differ from actual systems, requiring simplification compared to natural clusters, their strength lies in the ease with which they can be translated into the physical systems of practical design. This simplification, while departing from the intricacies of real-world animal clusters, enhances the model's power, rendering it highly effective for application in practical systems.

As discussed above, the existing models can be divided into bottom-up and top-down models. On the macro level, such classification includes almost all models and is comprehensive. On the micro level, however, such classification is less specific. Because it just focuses on the modeling method itself and ignores the specific mechanism in the different collective behavior, a more detailed classification is needed.

Firstly, we organized typical models based on their chronological emergence, starting from time when the most representative Boids model appeared up to the present. We found that the emergence of representative models during different periods allowed us to divide our focus timeline into four distinct periods. We summarized the common characteristics of representative models within each period and classified the existing typical models into four categories: models grounded in distance rules, models shaped by pheromone interaction, models incorporating hierarchical structures, and models stemming from empirical research. These categories are named as: self-driven particle models, pheromone communication models, leadership decision models and empirical research models. The categorization is illustrated in Fig. 3.[15–19,30,62–71] Unlike traditional bottom-up and top-down categorizations, our classification offers a more comprehensive perspective by incorporating the chronological sequence of each model type and their highlights shared characteristics over different periods.

3. Self-driven particle models

Self-driven particle models are the most classic models to describe SI. This kind of model simulates the individual in the biological cluster as a particle in a computer and realizes the movement behavior of a particle swarm by giving the particle a certain speed and designing simple local control rules, to simulate the simple collective behavior of the real biological group. Although this model is simple, many researchers have paid attention to it, as it is extremely effective.

3.1. Typical self-driven particle models

3.1.1. Boids model

The earliest and most classic model of self-driven particle models is the Boids model built by Reynold[15] in 1987 to simulate the movement of bird flocks on the computer. The most iconic feature of this model is that it provides three principles for simulating cluster movement: separation, alignment, and cohesion. The three principles are of decreasing priority. Separation means that individuals in a flock should keep a distance from their neighbors to avoid collisions. Alignment means that the speed of individuals should match the speed of individuals in the neighborhood as much as possible. Proximity means that individuals try to stay close to their neighbors. Reynold used these three principles, combined with distributed control and particle systems, to simulate the flight of birds in a computer.

3.1.2. Swarm model

Based on Boids model, the randomness of individual particle evolution was added to create the swarm model in Ref. 16. Particle velocity and acceleration in this model are described as

$$\boldsymbol{V} = c_1\boldsymbol{V}_1 + c_2\boldsymbol{V}_2 + c_3\boldsymbol{V}_3 + c_4\boldsymbol{V}_4 + c_5\boldsymbol{V}_5 \tag{1}$$

$$\boldsymbol{A} = \boldsymbol{m}\left(\frac{\boldsymbol{V}}{|\boldsymbol{V}|}\right) \tag{2}$$

where $c_i(i = 1, 2, 3, 4, 5)$ is a constant, $\boldsymbol{V}_1$ is the vector away from the neighbor within the crowding radius, $\boldsymbol{V}_2$ is a vector pointing to the center of the environment, $\boldsymbol{V}_3$ is the average neighbor speed of the particle, $\boldsymbol{V}_4$ is the vector pointing to the center of mass of all particles, $\boldsymbol{V}_5$ is a random vector, $|\boldsymbol{V}|$ represents the modulus of $\boldsymbol{V}$, and m is a constant that determines the maximum acceleration of the particle. Eq. (2) normalizes the velocity and sets the acceleration of the particle as the product of this result and m. By setting different parameters in

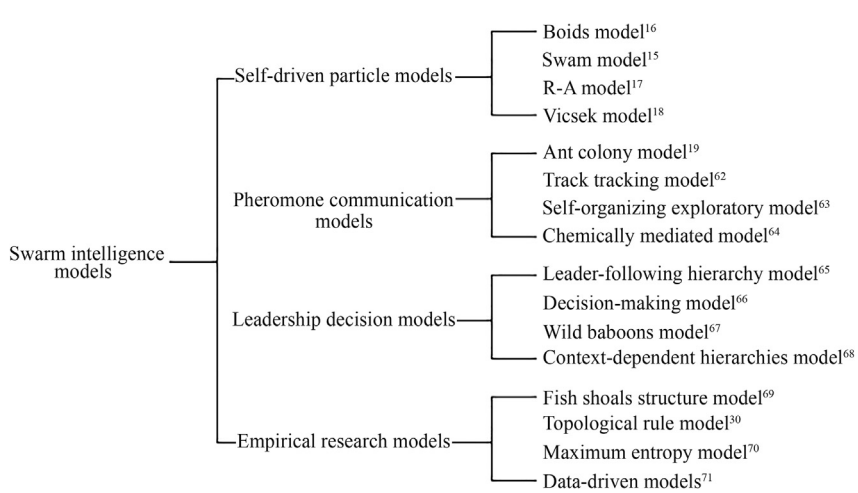

Fig. 3 A categorization of swarm intelligence models and their representative works.

the swarm model, different forms of swarm formation such as ring and figure-of-eight were identified in Ref. 72.

3.1.3. R-A model

The Boids model and swarm model are the simplest models of SI. Although they can simulate the move ment state of a swarm, these models cannot fully explain the well-defined collective behavior seen in many species. Based on the abstract description and the Boids model of an obvious aggregation tendency in biological systems, a more biologically realistic collective behavior model was established in Ref. 17, which is called the R-A model. It controls the cluster based on the following two rules:

(1) Individuals in the group try to keep a minimum distance from other individuals at all times, and this rule has the highest priority;
(2) If an individual is not equipped to implement rule (1), an individual will tend to gravitate toward others and align with its neighbors to avoid isolation.

Based on these two rules, the individual-centered surrounding area is divided into three non-overlapping spatial layers, as shown in Fig. 4,[17] which is a representation of an individual in the model centered at the origin: ZOR is the Zone of Repulsion, ZOO is the Zone of Orientation, and ZOA is the Zone of Attraction. The possible blind volume behind an individual is also shown. $\boldsymbol{\alpha}$ is the field of perception.

R-A model can simulate collective behavior based on the above rules. Because the constructed hierarchical space is 3D, it can also provide some explanation for the collective behavior of airborne biological clusters, and it is more realistic than the Boids model.

3.1.4. Vicsek model

Among the numerous self-driven particle models established from a physical perspective, the most representative model is Vicsek model developed in 1995.[18] The velocity of the particles in the model is determined by a simple rule and random perturbation. The only rule of the model is that at each time step, a given particle moving at a constant absolute velocity, its direction of motion is updated to the average direction of its neighbors, whose radius is r, and some random perturbation is added. In detail, at $t = 0$, N particles are randomly distributed in a square cell with the linear size of L, each particle has the same velocity v, but the direction θ is randomly distributed. The absolute velocity of all particles is measured at the same time in each period, and the position of particles is changed according to the following formula:

$$\boldsymbol{x}_i(t+1) = \boldsymbol{x}_i(t) + \boldsymbol{v}_i(t)\Delta t \tag{3}$$

where $\boldsymbol{x}_i(t)$ is the position of i at t, $\boldsymbol{v}_i(t)$ is the speed of i at t, and Δt is the unit time step. The velocity $\boldsymbol{v}_i(t+1)$ of i at $t+1$ is constructed as a vector whose absolute magnitude is v and whose direction is determined by $\theta(t+1)$, which is obtained by

$$\theta(t+1) = \theta(t)_r + \Delta\theta \tag{4}$$

where $\Delta\theta$ is the noise that is a random number selected from $[-\eta/2, \eta/2]$ with uniform probability, η is noise intensity, and $\theta(t)_r$ is the average value of the motion direction of surrounding particles, including the particle, in the neighborhood with a radius of r, expressed by an arctangent function:

$$\theta(t)_r = \arctan\left[\sin\left(\theta(t)\right)_r / \cos(\theta(t)_r)\right] \tag{5}$$

When $\mathbf{v}$, $\boldsymbol{\eta}$ and density $\rho = N/L^2$ are given, the simulation of different forms of SI can be realized based on the individual velocity and position updating rules. It is not difficult to see that this model is also simple, but it has a significant difference from the Boids model in that there is only one rule in this model, namely the alignment rule. The reason why the simulation of SI can be realized is greatly related to the selection of noise. The Vicsek model mimics the movement of a colony behavior model based on simple local alignment rules, but a theoretical explanation for this model is not provided. To this end, Jadbabaie et al.[73] analyzed the system properties of the Vicsek model from the perspectives of control and mathematics, gave its theoretical explanation, proved that it is a stable linear switching system, and pointed out that as long as the interactive network can remain continuous in time, the collective behavior can be realized.

Because of its simplicity and interpretability, the Vicsek model has been favored by many researchers, and some improved models have been produced. These models not only inherit the simple alignment rules but also add some more complex functions. For example, phase transitions in the collective behavior of local interactions of self-driven particles with cohesive and non-cohesive group noise were studied, and it was found in Ref. 74 that such transitions are always discontinuous. Gönci et al.[75] studied phase transitions in a 3D Vicsek model and found that the results are similar to those in a 2D environment. The 2D and 3D types of the Vicsek model were comprehensively studied in Ref. 76.

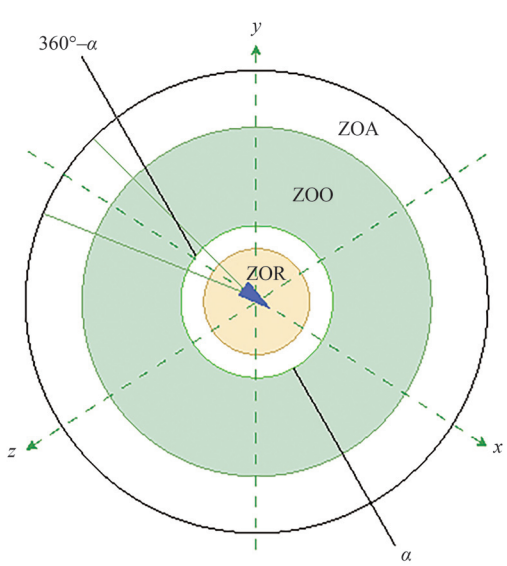

Fig. 4 Spatial stratification of R-A model.[17]

3.2. Particle swarm optimization

The models discussed above, in which the simulation of SI is realized, are typical self-driven particle models. In this section, PSO inspired by the social behavior of birds flocking or fish schooling, where each individual, called a particle, moves in

the search space following the best performance of its own and its neighbors, will be introduced. The mechanism of PSO is derived from the self-driven particle models, i.e., by simulating interactions among individuals within the population to optimize the problem. To some extent, PSO can be considered as a particular instance of self-driven particle models primarily used for optimization problems. On the other hand, self-driven particle models are more versatile and applied to simulate various self-driven systems and complex dynamic systems.

PSO was proposed by Kennedy and Eberhart[77] in 1995. In PSO, particles are updated iteratively by adjusting their velocity based on their own best position and the best position of their neighbors:

$$\boldsymbol{v}_{i,d}^{t+1} = \omega \boldsymbol{v}_{i,d}^{t} + c_1 r_1 (\text{pbest}_{i,d} - \boldsymbol{x}_{i,d}^{t}) + c_2 r_2 (\text{gbest}_d - \boldsymbol{x}_{i,d}^{t}) \tag{6}$$

where $\boldsymbol{v}_{i,d}^{t+1}$ represents the updated velocity of the i- th particle in the d- th dimension; $\boldsymbol{x}_{i,d}^{t}$ represents the position of the i- th particle in the d- th dimension; $\mathbf{pbest}_{i,d}$ represents the best position of the i- th particle in the d- th dimension; $\mathbf{gbest}_d$ represents the best position of all particles in the d- th dimension; ω is the inertia weight that controls the inertia of particle velocity; c_1 and c_2 are the learning factors that control the influence of particle's own and global best solutions; r_1 and r_2 are randomly generated values within the range $[0, 1]$ that are used to increase the randomness and diversity of the search.

In PSO, each particle maintains its current position and velocity, and continuously updates them to search for the optimal solution. The values of inertia weight, learning factors and random numbers in the velocity update formula need to be adjusted based on the specific problem. The pseudocode is shown as Algorithm 1,

Algorithm 1. Algorithm 1 PSO

```
for each particle i
    Initialize velocity v_i and position x_i for particle i
    Evaluate particle i and set pBest_i = x_i
end for
gBest = min (pBest_i)
while not stop
    for i = 1 to N
        Update the velocity and position of particle i
        Evaluate particle i
        if fit(x_i) < fit(pBest_i)
            pBest_i = x_i;
        if fit(pBest_i) < fit(gBest)
            gBest = pBest_i;
    end for
end while
print gBest
```

where N denotes the number of particles, $\mathbf{pbest}_i$ denotes the best solution of each particle, **gBest** denotes the global best solution, and fit represents fitness.

There are many enhancement approaches for the PSO algorithm structure, which can be classified into 10 main subsections shown in Table 1,[78–88] where the approaches in the first column represent the modifying method and the purpose in the second column clarifies the desired performance of the modified PSO.

3.3. Application of self-driven particle models

The main idea of self-driven particle models is to realize the control of collective behavior by using simple local interaction rules. Its simplicity and universality have attracted many researchers to study its application. Self-driven particle models have been widely used in the fields of robotics,[89] artificial intelligence,[62] complex systems[63] and control systems.[64] Then, we are focusing on multi-agent swarm control and the application of PSO, which can directly reflect the intrinsic idea embedded within the models.

3.3.1. Multi-agent swarm control

Multi-agent swarm control is to design the optimization algorithms to drive the behavior of a group of agents, called a swarm, to achieve a specific goal. Because of the simplicity of the three roles of the Boids model and the similarity between the swarm of animals and the multi-agent swarm, it has a wide range of applications in multi-agent swarm control.[90,91] In Ref. 92, the Boids model was used to character the behavior of the agents in the swarm. Specifically, the movement vectors corresponding to the repulsion and attraction force in the Boids model were used to design the movement algorithm of the shepherd, which exerts repulsive forces on the agents in the swarm to achieve spatial separation of a specified target agent from all the other agents, while maintaining the connectivity among the other agents. Figs. 5(a)–(d)[92] shows that a target sheep loses all the connections to the other sheep under the control of the shepherd.

Based on the swarm model, the interaction between an evolving swarm of birds and an environmental feedback controller using a simulation model was studied in Ref. 93. The simulation involves a flock of birds with localized energy sources that provide birds with the energy necessary for their survival. The model incorporates natural selection through an energy system in which birds die and are replaced by the offspring of the remaining population. Behaviors of agents evolve over successive generations to favor the most successful. However, due to the unicity of swarm model, the application of it in swarm control is not extensive.

The R-A model has a wide range of applications in multi-agent swarm control, particularly in flocking control, a subset of swarm control. It focuses on the behavior of a group of agents that move together in a self-organizing fashion, similar to the behavior of flocks of birds or schools of fish. In Ref. 94, the authors proposed a modified version of the R-A model based on fuzzy rules. The modified model improves the convergence probability and relative converging size of the flocking behavior. By inferring the R-A model, the researchers have developed a scheme for autonomous artificial agents to extract knowledge about the interactions within a swarm of interest.[95] They have done this by infiltrating the swarm of interest with a monitoring robot and constantly measuring the distance between the infiltrator and its nearest neighbor, which allows them to estimate the

Table 1 Enhancement approaches for PSO algorithm structure.

Approach	Purpose	Reference
Incorporating the utilization of multi-sub-population strategies	Enhancing diversity and search ability	79
Enhancing the selection strategy for particle learning objects	Accelerating convergence velocity	80
Modifying particle's update strategy	Preventing entrapment in local optima and enhancing its capability to navigate high-dimensional search space	81
Modifying velocity update strategy	Preserving diversity for comprehensive global search	82
Adjusting the speed or position constraint method and dynamically determining the search space	Dynamically determine the search space	83
Combining PSO with other search techniques	Increasing the diversity and improving the local search ability	84
Improving for multi-modal problems	Finding several better solutions	85
Keeping diversity of population	Improving the global convergence speed	86
Discrete PSO algorithm	Fitting discrete optimization domains	87
Multi-objective optimization PSO algorithm	Fitting multi-object optimization problems	88

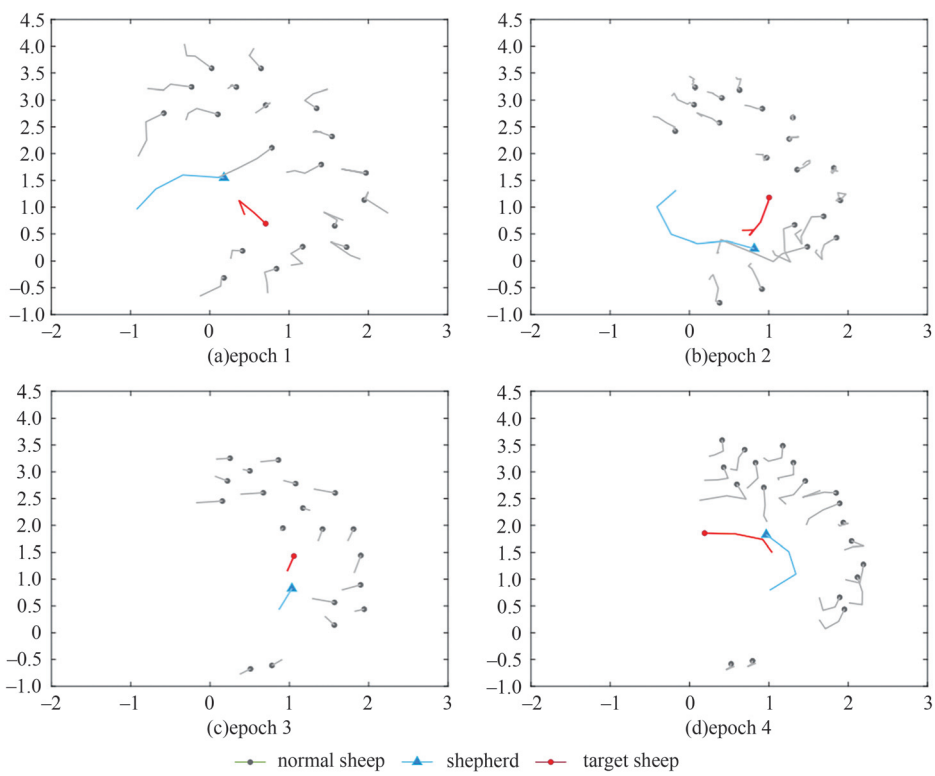

Fig. 5 A target sheep loses all connections to the other sheep under control from shepherd.[92]

repulsion radius of the swarm agents. This inferred model could then be used to control these swarms, for example, to guide them to safe areas. However, it is important to note that this method is limited to a specific model of interaction.

To achieve accurate UAV swarm navigation when only one UAV has predetermined path information, Tang proposed an improved Vicsek model with hierarchical weighting in Ref. 96. In Ref. 97, the Vicsek model has been used to analyze the behavior of swarm robots and establish a model of swarm robot synchronization. Self-organization regulations were proposed to achieve flock centering, direction coincidence, and collision avoidance. The simulation experiments showed that the proposed strategy can achieve swarm robot synchronization.

The thought of local interaction in self-driven particle models promotes the research in multi-agent swarm control. Berlinger et al.[98] demonstrated 3D collective behaviors with a swarm of fish-inspired miniature underwater robots that use only local interaction. Fig. 6 [98] demonstrates self-organization in space with (A) a shoal of surgeonfish foraging in a reef and (B) a dispersed black robot making its next move (black vector) based on a weighted average of attractive (blue) and repulsive (red) forces from nearby robots. (C) The inter-robot forces (red) are derived from the Lennard- Jones potential (blue) with standard parameters $a = 12$ and $b = 6$, and a tunable target distance $d_t = 2$ BL (Body Length), which defines the robot density of the group. The forces f depend on the distance d between robots: $f = 0$

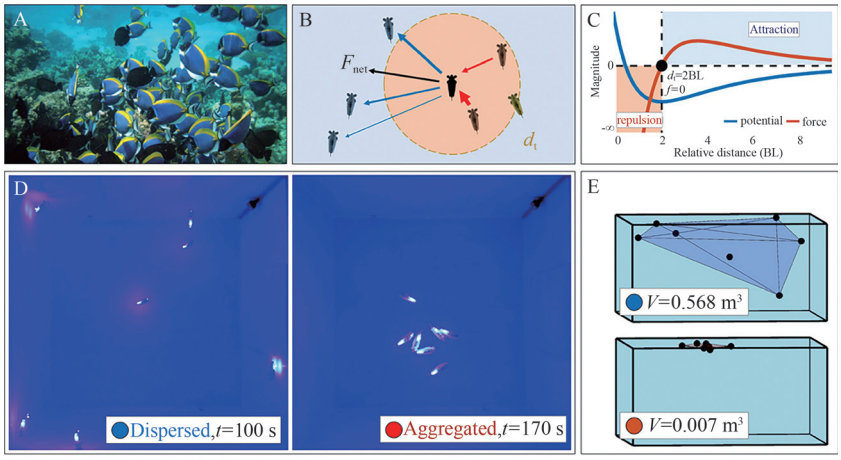

Fig. 6 Self-organization in space.[98]

when $d = d_t$, $f \ll 0$ when $d \ll d_t$ (repulsive), and $f > 0$ when $d > d_t$ (attractive). (D) shows dispersion (blue markers, $d_t = 2$ BL) and aggregation (red markers, $d_t = 0.75$ BL) with seven robots. (E) shows the volume of schools in two situations. Moreover, based on the distribution strategy in biotic populations, the synchronization in multi-agent systems, as a part of swarm control, has been well developed. Ref. 99 overviewed the synchronization in multi-agent system comprehensively.

3.3.2. Applications of PSO

PSO has been successfully applied to solve a wide range of optimization problems in various fields, especially aerospace area. This section focuses on its application in the aerospace area, in which aircraft design and flight control will be narrated. In aircraft design, PSO can be used to optimize aerodynamic performance, structural design and weight distribution of the airplane to improve flight efficiency and economy. For example, PSO can be used to design the airfoil,[99] fuselage profile[100] and engine layout of anairplane.[101] Especially, In Ref. 102, a longitudinal parameter identification procedure for a small UAV through modified PSO was described. The usefulness of this method was demonstrated by experimental validation.

In flight control of the airplane, it can be applied to the parameter adjustment of the flight control system, such as PID controller parameter optimization in the autopilot system. Besides, it can also be used to complete active disturbance rejection control for aircraft anti-skid braking systems. In Ref. 103, an improved version of PSO algorithm was used to adjust the parameters of the anti-skid braking active disturbance rejection controller, which helps to improve the performance and safety of the aircraft braking system. In addition to the above applications, various applications of PSO can be found in Ref. 78, which is a comprehensive review of PSO.

3.4. Evaluation of self-driven particle models

There is no doubt that the self-driven particle models achieve the description of SI through simple local control rules. At present, most of the self-driven particle models focus on the study of the same population of organisms, and there are few studies on heterogeneous populations. It is a feasible method to model the collective behavior of heterogeneous populations by setting different types of particles in the system to simulate the collective behavior of different populations. Therefore, the future research direction of self-driven particle models should focus on the research of heterogeneous population collective behavior models. Additionally, these models are also unauthentic and cannot character some more basic collective behaviors of organisms in nature because they are too simplistic, for example, how ant colonies move food. There is, however, a type of model that does a good job of explaining the mechanism of SI of ant colonies and other similar animals. These models model a remarkable feature of these animals: they all release pheromones for group communication when they cluster.

4. Pheromone communication models

Pheromone communication models are a type of model used to explain the specific interaction patterns within certain biological clusters, in which individuals communicate with each other through pheromones unique to the species. It should be noted that this differs from the Physics-Chemistry-Mathematics-based (PCM) algorithms, which typically rely on mathematical, physical, or chemical principles to solve problems or model natural systems. Pheromone communication models, on the other hand, are based on the concept of pheromone communication. There are many pheromone communication models, such as ant colony models[19] in which ants find the best path through pheromone communication, bee colony models[104] in which worker bees guide other bees to find nectar sources through pheromone communication, and bacteria foraging models[34] in which coordination in the search process is achieved through pheromone communication. For the sake of brevity, this article focuses on representative models of pheromones in ant colonies.

4.1. Typical pheromone communication models

4.1.1. Ant colony path optimization model based on pheromone

The research on ant colony collective behavior started from human observation that ants can always find the shortest path

when transporting food. Through the observation of ants transporting food, people found that ants would leave pheromones on the path, but it was not clear how ants find the shortest path mechanism. In 1989, by designing a simple experiment, Goss et al.[19] studied the mechanism of ant colony to find the shortest path through collective cooperation, and established model to explain the mechanism of ant colony to find the shortest path. In this model, ants pass the path shown in Fig. 7 [19] every second from each direction and place one pheromone unit in each direction. When a choice point $j = 1, 2$ is reached in the model, each ant takes a long or short path with a probability of $P_{L,j}$ and $P_{s,j}$, respectively, depending on the pheromone content S_j and L_j of the two branches of the choice point and then adds a pheromone to the chosen branch.

Based on their average speed and the length of the short branch, those ants that choose the short branch at $j = 1$ (or 2) will arrive at the opposite choice point $j = 2$ (or 1) about seconds later, and add pheromones there. Those who chose the long branch would do the same thing $20r$ seconds later, where r is the ratio of the length of the long branch to the length of the short branch. The timescales of these experiments were on the order of magnitude of the average life span of the pheromone,[20] to avoid the error caused by the evaporation of the pheromone.

Therefore, at time t, the average equations of the random system are:

$$\frac{\mathrm{d}S_j}{\mathrm{d}t} = P_{s,j'}(t-20) + P_{s,j}(t); j = 1, j' = 2 \text{ or } j = 2, j' = 1) \tag{7}$$

$$\frac{\mathrm{d}L_j}{\mathrm{d}t} = P_{L,j'}(t-20r) + P_{L,j}(t); j = 1, j' = 2 \text{ or } j = 2, j' = 1 \tag{8}$$

$$P_{s,j} = \frac{(20+S_j)^2}{(20+S_j)^2 + (20+L_j)^2}; P_{s,j} + P_{L,j} = 1 \tag{9}$$

where Eq. (9) is the selection function obtained according to the experiment in Ref. 21. As the path would increase gradually with time, it was concluded that the ant's shortest path-finding method is closely related to the pheromone. The model is a random system model, which accurately quantifies the pheromone and makes the ant's choice probabilistic. Moreover, the model's explanation of the ant's finding the optimal path is highly consistent with the observed phenomenon, which is a simple but effective model.

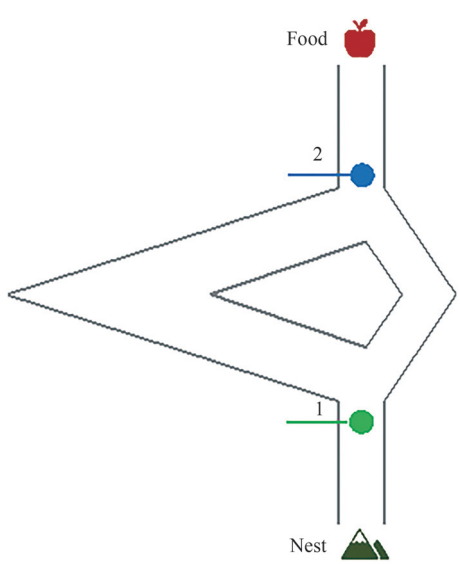

Fig. 7 Experimental path of ant colony collective behavior.[19]

4.1.2. Influence of parameters on pheromone communication model

The foregoing model has fully revealed a mechanism of ants choosing the shortest path, which is the result of the continuous synergistic action of ant colonies. Every ant chooses different branches according to the concentration of pheromones, and at the same time leaves new pheromones for the next individual to recognize. In this way, there is a circular interaction between the groups, forming a positive feedback effect between the groups, and finally enabling the ant colony to carry food along a shorter path. This simple but effective collective behavior is so impressive that many researchers have been devoted in this study to reveal the mechanism of ant colony collective behavior, and to study the influence of specific parameters in the model.

Focusing on the transition between colony foraging and solitary foraging, a track-tracking model that combines biological parameters associated with organisms, tracking markers, and populations was established in Ref. 105. Ordinary differential equations relate to the density of branches and the number of individuals tracked or randomly explored by track-tracking. Models reveal that subtle changes in individual behavior can lead to large differences in group behavior, including the ability to switch from individual actions to organized collective behavior. Compared with the models in Ref. 19, this model paid more attention to the state transition of individual ant colonies during the whole foraging process, revealing the mechanism of food transport route selection of ant colonies in more details. In Ref. [106–109], the spatial structure and colony behavior of ant colonies were simulated. In some simulations, deposition and evaporation of pheromones affect the ant's local direction and speed of movement.

4.2. Ant colony optimization

The SI of ant colonies seeking the shortest path was cleverly exploited by Dorigo.[110] Based on the mechanism of finding the shortest path in ant colonies, he put forward the ACO, which is the basis of the ant colony pheromone model that can achieve a wide range of applications. Its pseudocode is shown as Algorithm 2.

Algorithm 2. Algorithm 2 ACO

```
Initialize ant colony
Initialize pheromone matrix
while (termination not met):
    for each ant:
        Choose next node
        Update pheromone levels
    end for
end while
```

In line 5, the next node is chosen by

$$p_{ij} = \frac{[\tau_{ij}(t)]^{\lambda} \times [\eta_{ij}]^{\xi}}{\sum_{k \in \text{allowed}} [\tau_{ik}(t)]^{\lambda} \times [\eta_{ik}]^{\xi}} \tag{10}$$

where p_{ij} represents the transition probability from node i to node j; $\tau_{ij}(t)$ represents the pheromone concentration on the path from node i to node j at time t; η_{ij} represents the heuristic information from node i to node j; λ and ξ are the importance factors of pheromone concentration and heuristic information, respectively; J_i represents the set of all unvisited nodes. And at line 6, pheromone levels are updated by

$$\tau_{ij}(t+1) = (1-\gamma) \times \tau_{ij}(t) + \sum_{k=1}^{m} \Delta\tau_{ij}^{k}(t) \tag{11}$$

where γ is the pheromone evaporation factor; $\Delta\tau_{ij}^{k}(t)$ is the amount of pheromone released on the path by ant k at time t.

The essence of ACO is to realize the information interaction between groups based on pheromone, and then complete the search and optimization process. In the following period, various improved algorithms appeared one after another, which improved the theoretical research of ant colony algorithm and promoted the corresponding application research, including the typical Ant Colony System (ACS)[111,112] and Max-Min Ant System (MMAS).[113,114] The related improved algorithms are summarized in Table 2,[66,67,113,115–121] where TSP represents Traveling Salesman Problem.

4.3. Application of pheromone communication models

The main applications of pheromone communication models are related to ACO. It is a powerful metaheuristic algorithm inspired by the foraging behavior of ants. With further research, it has been found that ACO has significant advantages in the field of path planning and obstacle avoidance. On the one hand, these problems themselves are high dimensional, nonlinear, and highly complex problems, while ACO has good search and optimization capabilities, making it well-suited to solve these types of problems. On the other hand, these problems also have important practical applications, such as autonomous driving, drone navigation, and robot control, making research in these two application areas of practical significance. In this article, we will explore the applications of ACO in path planning and obstacle avoidance.

Table 2 Overview of main ACO algorithms for NP-hard problems.

ACO algorithm	Reference	Year	TSP
Ant system	116	1991	Yes
Elitist AS	117	1992	Yes
Ant-Q	118	1995	Yes
Ant colony system	65	1996	Yes
MMAS	113	1996	Yes
Rank-based AS	66	1997	Yes
ANTS	119	1998	No
Best-worst AS	67	2000	Yes
Population-based ACO	120	2002	Yes
Beam-ACO	121	2004	No

ACO has been used for path planning and obstacle avoidance, and has demonstrated effective performance in complex and dynamic environments. For example, in Ref. 122, an improved ACO by introducing the yaw angle was used to generate a path for a UAV in low-altitude penetration. In another study in Ref. 123, an improved ACO using the characteristics of the A* algorithm and MMAS was applied to optimize the path of a mobile robot to avoid obstacles in real time. To improve the performance of ACO-based path planning and obstacle avoidance, researchers have developed several techniques, such as an adaptive heuristic function and hybrid algorithms. An adaptive heuristic function has been proposed to address issues like low search efficiency, slow convergence speed, and getting stuck in the local optimum.[68] This function updates heuristic information based on the shortest actual distance that an ant has traveled. Hybrid algorithms that combine ACO with other optimization techniques were proposed to improve the efficiency of the search process.[124]

Despite the success of ACO-based approaches for path planning and obstacle avoidance, there are several challenges that must be addressed. One such challenge is dealing with dynamic environments, where obstacles may appear or disappear during the search process. To overcome this challenge, researchers have proposed techniques such as dynamic pheromone updating[125] and moving obstacle avoidance.[126] Another challenge is the need to balance exploration and exploitation during the search process. To address this, researchers have developed techniques such as multi-objective ACO[127] and parallel ACO.[128] Multi-objective optimization seeks to optimize multiple objectives simultaneously, while parallel ACO distributes the search process among multiple agents to improve efficiency.

In particular, the basic ACO does not include the part that can make an agent avoid obstacles, so we summarized the related modifying method of ACO to avoid obstacles. The main method is combining ACO with other strategy. For example, in Ref. 129, a dynamic path planning algorithm, achieving collision avoidance through fuzzy logic control, was proposed. This algorithm enhances the ant colony system's pheromone update mechanism using a sigmoid gain function for efficient exploitation during path planning. In Ref. 130, a time taboo-based occupancy grid prediction model for handling unknown time-varying dynamic obstacles was proposed, which establishes an ant colony information inheritance mechanism. Another effective method is adding the penalty term in fitness function or objective function for avoiding obstacles, such as Ref. 131.

In summary, ACO is a promising approach for path planning and obstacle avoidance in robotics and autonomous systems. With the use of adaptive and hybrid algorithms, as well as techniques for dealing with dynamic environments and balancing exploration and exploitation, ACO has the potential to achieve efficient and effective navigation in complex and dynamic environments.

4.4. Evaluation of pheromone communication models

The SI model established based on pheromones fully demonstrates that the collective behavior ability of a colony can be hardwired to the components of the system, in which hardwired refers to the real physical medium, pheromone,

through which groups interact to realize the movement between groups.

Pheromone communication models are close to nature. If the main differential equation of pheromone communication models is established properly, and the natural properties of network structure, path size, individual heterogeneity and population are modeled carefully, the model will be more realistic for simulating the collective behavior based on the pheromone. Although the research on the pheromone communication models has been relatively sound, in the specific application field, it mainly focuses on the ant colony pheromone model, and there is a lack of application research on other SI models. Therefore, the future development of pheromone communication models should focus on application and explore the application fields of pheromone models outside ant colonies.

5. Leadership decision models

In some biological clusters, individuals interact with each other not through distance rules but based on topological rules, such as starling flocks. In the actual study, the researchers found that they interact not just based on topological rules, but also based on hierarchies. For example, in flying of starling flocks, they arrange themselves in different formations, and individuals in different positions in the formation have different hierarchies. In fact, hierarchies exist not only among groups of birds, but also among groups of other organisms, such as wolves, baboons, and sheep. Because of the existence of leadership mechanism, the interaction and decision-making between groups are no longer based on simple topology or distance rules. Therefore, if the established model can simulate and explain the collective behavior of these groups well, it is necessary to establish a model based on the leadership and decision-making mechanism, and it is particularly important to discover the leadership and decision-making mechanism among biological clusters.

5.1. Typical leadership decision models

Group decision-making is a complex and varied process, with information sharing ranging from total openness to complete secrecy. Figuring out exactly how this process works can be tough, but it's crucial to understand how groups make decisions based on their leadership structures.[132] Which part of the group or which individual is the leader needs to be defined. Only by finding the leader can we study the decision-making process. However, the definition of a leader generally uses the feature of influence. If an individual's behavior causes certain behavior changes in other individuals in the group, this individual has influence; if an individual has repeated influence on the behavior of other individuals, whether directly or indirectly caused by the hierarchy, this individual will become a leader.[133]

Based on the definition of a leader, the researchers studied the internal information transfer mechanism and decision-making mechanism when a group performs collective behavior. In Ref. 134, Global Position System (GPS) was used to acquire 3D data, the leader–follower hierarchy model of pigeon flocks was studied, and the hierarchy dynamics model of pigeon flocks was established. Based on the Boids model, the mechanism of information transfer and leadership decision-making in the biota were studied in Ref. 135. The model showed that effective information transfer and decision-making can occur in animal groups without a clear signal or information transfer mechanism. It has been demonstrated in Ref. 136, both theoretically and experimentally, that in many cases an opinionated minority of individuals can determine group selection, but the presence of ignorant individuals will spontaneously inhibit this process, returning control to the numerical majority group of individuals. The results highlight the role of ignorant individuals in achieving democratic consensus in situations of internal group conflict and information constraints. The study in Ref. 137 tracked wild baboons with high-resolution global positioning systems and analyzed their relative movements to each other, revealing a process of shared decision-making that governs baboon movements. Rather than preferentially following the dominant individual, baboons are more likely to follow multiple initiators. When there was a conflict in the direction of movement, the baboons would choose the direction with the greater angle, but they would compromise if the angle was smaller. These results are consistent with models of collective movement, suggesting that democratic collective behavior that emerges from simple rules is universal, even in complex, socially stratified societies, which likewise validates the theory in Ref. 137. Thus, an individual with a wide range of social connections is more likely to influence group decision-making. Instead, in some cases, influence may be a byproduct of social connections rather than a direct consequence.

Since most animals have visual senses, the communication process of decision-making can also be based on it. It was suggested that an individual's motor decisions are explicitly based on the sensory information available to the organism in Ref. 138. In other words, they argued that while spatial relationships do affect interactions between individuals, they do so indirectly through the individual's detection of sensory cues. In the experiment, they used the data to reconstruct the visual field of each individual, based on which they analyzed the information transmission mechanism and decision-making mechanism of fish shoals, and explicitly considered that visual perception could more accurately predict the propagation process of behavioral change in groups with a leadership hierarchy. Based on the assumption of visual perception, Rosenthal et al.[139] studied the hidden information transmission networks in the group, revealed the nature of decision transmission, complex and local, by studying these interactive networks, and determined that individuals with relatively few but closely connected neighbors are both the most socially influential and the most vulnerable to social influence.

5.2. Summary of leadership decision models building methods

The determination of leaders and the decision-making process depends on the influence of individuals on individuals and the influence of individuals on the group. A methodological framework for group leadership decision-making mechanisms based on influence research was proposed in Ref. 133, which is used to guide the acquisition and interpretation of experimental data. The framework can be divided into three steps. The first step, which identifies the situations in which individuals exert influence over the rest of the individuals in one instance, can be divided into two small steps. First, the level of individual-

to-individual influence can be studied, i.e., who influences whom in a given group decision-making process, and the potential strength of these influences; second, the impact of individuals on the group level is studied, i.e., quantify the overall impact of individuals on the group level. The second step, quantifying the overall impact, is to summarize the impact of the first step through repeated events and quantify the impact of the individual on multiple group decisions. The third step, judging the consistency of the influence, is to determine whether the influence of an individual on other individuals in the group varies over time or with the decision event.

With the development of computer technology and tracking techniques, methods of inferring influence have also been adequately developed. Specifically, these methods can be used to sort out the methodological framework of group leadership decision-making mechanism based on influence research, which can be summarized as the methods of inferring the individual-to-individual influence in an instance and the individual-to-group influence based on observed data, the methods of inferring the overall influence, and the methods of assessing the consistency of influence. The above methods and their advantages and disadvantages are summarized in Tables 3–5.[69,70,140–149]

5.3. Application of leadership decision models

This biological cluster with leadership level has a strict hierarchical distribution when producing collective behavior, so the establishment of the model is also carried out around the discovery of the leader in the group. The above framework completely describes the basic modeling idea for the behavior of an organism cluster with a leadership decision-making mechanism. Through a variety of model studies, people have become more and more clear about the leadership decision-making mechanism of SI, and many application examples based on this have been produced.

The application of the leadership decision models is mainly realized by the leader–follower strategy. Because the leadership decision models are based on the formation phenomenon of birds and other biological clusters in motion, the leader–follower strategy is mainly used in formation control.

In formation control, the basic structure of leader–follower strategy includes the leader and the follower, and the leader is responsible for planning and compiling the team line, where the follower maintains an ideal formation by controlling their position relative to the leader.[150] In Ref. 151, an improved leader–follower formation control algorithm combined with Kal-

Table 3 Methods for inferring influence of individuals on individuals and groups.

Method of inferring effect in instance	Advantage	Disadvantage	Reference
Spatial location inference method	Can be directly assessed directly through observation	Assumptions may be invalid.	140
Initiator-follower identification method	Broad applicability	Inapplicable when individual distances change	141
Delay time inference method	Robust scalability	Unsuitable for non-sustained movement	142
Entropy measures	A solid and widely applicable theoretical foundation	Lacking a basic model when the interpretation is low	143
Decision result regression method	It is both explanatory and easy to analyze	Inapplicable without knowledge of individual preferences	144

Table 4 Methods of inferring aggregate impact.

Method of inferring aggregate effect	Advantage	Disadvantage	Reference
Influence Elo method	Wide applicability	Results lack significance with a large number	145
Social network inference method	User-friendly operation	Overall situation comprehension is challenging	146
Maximum entropy model method	Suitable for groups with large datasets	Computationally burdensome	69
Model fitting method	Applicable at both individual and group levels	Ensuring biological effectiveness is difficult	147

Table 5 Impact conformance assessment methods.

Method of assessing impact conformance	Advantage	Disadvantage	Reference
Standard index method	Broad applicability to independent data	Inapplicable to non-independent data	70
Randomity-based modeling	Suitable for non-independent data	Prone to dimensional disasters	148
Mantel test method	Facilitate hypothesis testing	Lack consideration of spatial autocorrelation	149
Coarse-grained evaluation method	Easy to analyze and manipulate	Result in low accuracy	142

man Filtering was proposed to address the problem of multi-robot formation control. The algorithm introduces virtual robots to assist in the formation and transformation of multi-robot and uses laser radar for planning and controlling. The leader robot acts as a reference point to guide the formation, while the following robot moves to the target through the guidance of the virtual navigator, forming the expected combination structure.

The leader–follower strategy was also used to realize formation control of UAVs.[152] Yang et al.[71] proposed a trajectory tracking control method for leader–follower quadrotor UAV formation, addressing the challenges of external disturbance, uncertain aerodynamic parameters, and collisions among formations. The control strategy utilizes a backstepping adaptive control approach for the leader UAV and formation sliding mode control for the follower UAVs. The effectiveness of the proposed method is demonstrated through simulation and experimental platforms, showcasing its potential for real-world applications in coordinated flight. The study in Ref. 153 addressed the problem that the number of leaders in a given position is not significant to determine the follower position in the traditional leader–follower strategy. Based on the motion vector information, the leader–follower strategy was improved to solve this problem.

Since the formation control is generally transformed into a consensus control problem, the leader–follower strategy is also used for multi-agent consensus control. In terms of multi-agent consensus control, the leader is usually independent of the followers, but it will have an impact on the followers. At this time, all the agents can be controlled as long as the state of the leader is controlled. In the realization of multi-agent consensus control with leader–follower strategy, the difficulty lies in the fact that not all followers can directly communicate with the leader. Based on the methods to deal with this difficulty, the existing researches on leader–follower consensus control can be roughly divided into three categories:[154]

(1) The leader's behavior is determined by a dynamic node that is similar to a follower with zero or known input.[155–159]
(2) Assume that the desired reference locus is linearly parameterized, and then all subsystems know the basis function vector.[160]
(3) Refer to the time-varying distributed control method using a non-smooth sign function, but it is not ideal due to the chattering phenomenon. [161]

The research in Ref. 154, aimed at the consensus problem, successfully relaxed the previous requirement that the leader's trajectory be linearly parameterized with some known functions, that is, relaxed the type (2) hypothesis above. In addition, the consistency problem of multi-agent systems with leaders under two conditions of fixed topology and switching topology was studied in Ref. 162, while the leader–follower consensus problem of multi-agent systems with invariant linear multi-agent systems at finite data rates was studied in Ref. 154.

In short, the application of the leadership decision models does not directly apply the mechanism similar to the SI, but realizes the specific application based on the idea in it, that is, to realize the overall control of the group behavior by controlling the behavior of a part of individuals. When the model is applied to the actual robot crowd, UAV swarm and unmanned boat swarm, it can usually play a role in reducing the communication consumption and control cost.

5.4. Evaluation of leadership decision models

The leadership decision models are established based on the hierarchical mechanism between biological clusters. The leadership hierarchy and decision-making process within the group are studied according to the hierarchical mechanism. Most of these models are built on the basis of experimental data, so the effect of the models is more realistic, and at the application level also has a very considerable development prospect. The future research direction should focus on applying the leadership decision mechanism existing in biological clusters to the decision planning of unmanned systems.

6. Empirical research models

The self-driven particle models are rule-based models, while the pheromone communication models are medium-based models. However, due to the relative underdevelopment of technology, the establishment of these two models lack empirical research to verify, and the rationality of their models is not convincing to some extent. However, since the development of advanced testing methods such as camera technology and laser radar, researchers have had the conditions for the empirical study of biological cluster behavior, and thus the SI has been studied at a deeper level. The establishment process of the leadership decision models above is actually similar to empirical research models, which require data-based modeling and analysis, involving the steps of data collection, data processing, model establishment, and model verification, but the analysis focuses on finding the formation mechanism of the leading decision in the group. What is introduced in this section is empirical research models except the leadership decision models. In fact, it is a kind of model based on advanced testing means and established through investigation. The ant colony model introduced above also relies on part of experimental data in the establishment process, but the difficulty of data collection and analysis is relatively low and does not involve the measurement of relative distance and the processing of complex data. Moreover, the experiment is artificially designed and does not occur naturally in biological clusters, so it is not the focus discussed in this section. This section mainly discusses models related to the behavior of biological clusters, which are relatively difficult to measure and calculate with actual data.

6.1. Overview of empirical research models

6.1.1. Small-scale cluster empirical research models

Since the model is established based on empirical research, the spatial structure it considers must be real 3D space. As long as the method adopted can observe and measure the research object in an all-round way and the data processing scale is within the feasibility, the empirical study of 3D biological cluster can be realized. To obtain 3D data, the simplest way is to use photography technology to obtain the relative position and relative distance of objects, and then use the actual acquired data for analysis and research. For example, the study on the 3D structure of fish shoals in Ref. 163 is a landmark study

on 3D biological clusters. In the study process, the author created the 3D method and the shadow method to measure the three-dimensional distance, which set a precedent for empirical research on 3D biological clusters. Ref. 164 quantified the three-dimensional trajectory of 8 fish using stereo photography and a computer tracking algorithm and compared the polarity, velocity, and average nearest distance of a real shoal with a simulated model. In Ref. 24, video recording and image digitization techniques were used to observe fish shoals, and evidence of attractive and repulsive regions was found. In addition, three novel group structure features were also found, and the nearest neighbor rule model of the self-driven particle models was verified by the empirical study.

6.1.2. Large-scale cluster empirical research models

Foregoing models are all research based on the data obtained from experiments. Some of them not only verify the rationality of the attraction and exclusion rules in the self-driven particle models but also discover new group structure characteristics in SI, which greatly promotes the development of the research on SI. However, the above-mentioned three models target a small population size, the number of individuals only maintained at around 10, and the conclusions obtained are not necessarily applicable to large-scale groups. This is because the proportion of individuals in the periphery of a large group is significantly lower than that in the center, but this does not happen in small groups. Therefore, for large groups, this marginal phenomenon is significant to cause the interaction rules of the group to be inconsistent with the three principles of the self-driven particle models. Therefore, new methods and techniques are needed to study the behavior of large-scale biological clusters.

By recording the whole process of SI of the population with the camera, and analyzing the recorded data later, the model established will have a higher similarity to the real collective behavior of organisms. Photos of hundreds of starlings during migration have been taken using high-speed cameras in Ref. 23. After analyzing the data in the photos, a topological rule model of starlings was established. This overturns the previous distance measurement rules, and the conclusion shows that the interaction between starlings is not based on distance range, but based on topological rules, and the specific number of interacting neighbors is 6–7 starlings around each individual. The study in Ref. 165 used a maximum entropy model to analyze starling flock data. The study found that local, paired interactions between birds were significant to predict the flock's order spread, and the number of interacting neighbors did not depend on population density. This supports the idea that interactions are determined by topological distance rather than physical distance. Ref. 166 reported a study on a large flock of starlings, which found that changes in the behavior of one bird affected the behavior of all other birds in the flock, regardless of flock size. This scale-free behavioral correlation suggests the flock is a critical system with a limit to its response to environmental disturbances.

The work in Refs. 23 and 166 actually relied on a project called STARFLAG, which studies the collective behavior of large clusters of organisms. The main technical issues in the collection of 3D data for large animal populations were reviewed in Ref. 167. In particular, it explains how to solve the stereoscopic matching problem, which has been a major bottleneck in all past 3D studies. It also shows how to avoid the pitfalls most common in the quantitative analysis of 3D animal populations, paying particular attention to the problem of bias introduced by the boundaries of a population, and illustrating by example the problem of artifacts resulting from ignoring boundary effects when studying the 3D structure of a population. In the course of the STARFLAG project, some methods are provided for studying the behavior of large clusters of organisms, and the boundary effect of the mentioned large clusters relative to small clusters is also taken into account in the study, and the results show that its influence on the behavior of biological clusters is significant.

6.2. Summary of methods for building empirical research models

New tracking techniques have made it much easier to collect data, and vast datasets have been uploaded to the Internet for researchers to use. The immediate problem is how to use such large data sets to construct more accurate models. Typical approaches to solve this problem are the Force Map and data-driven modeling methods.

Since the collected data sets are generally trajectory data about time series, which contains the movement direction and spatial position of individuals at a certain moment. The time difference method can be used to infer the movement speed, acceleration, relative position, and relative direction of individuals at a certain moment. For groups, the aggregation density and degree of polarization between groups can be counted to further infer the interaction force between individuals in the group. Force Map is a method for directly inferring the interaction forces between individuals from experimental data.[168,169] This method forms variable space with variables such as relative position and the relative direction of movement between individuals. By drawing the distribution of acceleration and angular acceleration in it, it can infer what kind of social force exists between individuals, and analyze the relationship between social force and relative motion state. In Ref. 147, Force Map was applied to analyze the trajectory data of golden shiners, and it was found that the social forces between individuals showed the effects of repulsion and attraction, and the directional alignment was the result of both effects. At the same time, the social forces from multiple neighbors did not act on individuals in an average manner.

The Force Map method can infer the social forces according to the trajectory, which is similar to the self-driven particle models, but it is actually verifying the rationality of some rules in the self-driven particle. Although simple in nature, it is at the cost of sacrificing important constraints and oversimplification. Four of its most critical constraints were summarized in Ref. 170. In order to overcome the limitations of the Force Map method, a data-driven modeling approach was proposed in Ref. 170, which can quantify and map the interaction between two individuals, describe how the strength of these interactions varies as a function of individual state variables, reconstruct the interaction function, and derive a clear and concise mathematical model to reproduce the observed behavior.

6.3. Application of empirical research models

As mentioned above, the key step of empirical research lies in the collection of data, which is most notably reflected in the

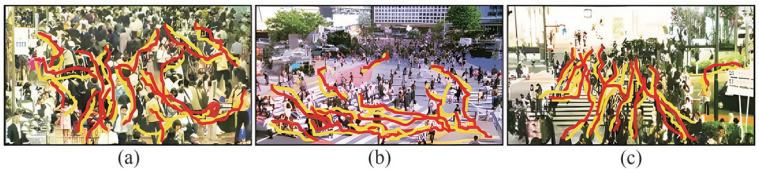

Fig. 8 Testing videos along with manually annotated ground truth (yellow) and data-driven tracking results (red).

data-driven model. The data- driven model not only verifies the correctness of the rules of the self-driven particle models, but also studies a new SI model on this basis. Nevertheless, the key role of empirical research models is to verify the correctness of the previously established models, referring to the self-driven particle model and other models based on observation and simulation. Therefore, the application of the empirical research models is mainly reflected in the application of its research idea, that is, the modeling and prediction of group behavior based on a large number of experimental data, or in other words, the data-driven application research.

6.3.1. Trajectory tracking in dense crowd based on data-driven method

The application of the data-driven method which is closely related to the behavior of biological clustering is reflected in the track tracking, among which the track of population trajectory is the most important application. The tracking of crowd trajectory here is for the consideration of public safety, which refers to the tracking of crowd trajectory in large and dense scenes. The tracking method is a tracking algorithm. The designed track tracking algorithm and monitoring system were used in Ref. 171 to realize track tracking for multiple individuals in a crowd in a large and dense scene. The accuracy of many pedestrian tracking algorithms is achieved by predicting pedestrian tracks through crowd motion models. A crowd motion model based on the local crowd motion modes in intensive videos was pro- posed in Ref. 172. This model encodes the time evolution of local motion modes represented by directional statistical distribution and then uses this model to estimate the expected movement of pedestrians at each spatiotemporal position, thus achieving a visual measurement of pedestrian efficiency, and using this pedestrian efficiency to detect abnormal events and track individuals in crowded scenes. The crowd motion tracking algorithm proposed in Ref. 173 is driven by prior behaviors, which are learned from a large number of crowd video databases collected from the Internet. Different from Ref. 172, this algorithm learns crowd motion modes based on the extracted global features. Fig. 8[173] shows the results of Ref. 173, where (a) is a busy airport scene, (b) is the Shibuya crosswalk, and (c) is a two-way zebra crossing scene. Combining the multi-agent motion model based on speed obstacle and high-definition crowd video data set, a real-time tracking algorithm for crowd trajectory tracking under medium density was proposed in Ref. 174.

6.3.2. Social network analysis

The empirical research models are computational models that simulate the behavior of biological swarms and can be used to analyze user behavior in social networks. In social network analysis, the biological swarm empirical research models can be used to simulate social interaction and information dissemination characteristics of users.

For example, it can be used to study information dissemination in social networks. The model can simulate information exchange between different users, as well as the speed and scope of information dissemination. In this way, factors influencing information dissemination can be identified and the most effective information dissemination strategies can be determined.

Additionally, it can be used to analyze users' social networks. By modeling the interactions and connections between users, the relationships between different users in the social network can be determined. These relationships can be used to predict information dissemination and influence between users. Focusing on identifying potential user groups that may play a role in the propagation of misinformation and information, Ref. 175 used K-means integrated with bio-inspired algorithms to analyze 10,000 users and 555,684 tweets to compute eighteen factors based on the tweet and user parameters. Factor selection for the final analysis identified 11 statistically relevant factors. In summary, the empirical research models can provide valuable information in social network analysis, helping us better understand and manage user behavior in social networks.

6.4. Evaluation of empirical research models

The empirical research models have been verified through real-world experiments, so compared with the self-driven particle models and pheromone communication models, their authenticity and accuracy are improved to an absolute degree, and it is more convincing in revealing the mechanism of SI. As can be seen from the above overview, the development of empirical research models is actually closely related to the development of data collection methods and data processing techniques. Therefore, the future development of empirical research models depends on the development of advanced data collection and processing techniques. At present, the empirical research models should be mainly developed with the new generation of large-scale data processing technology. For example, deep learning is introduced into the model parameter determination to realize the establishment of a large-scale collective behavior model.

7. Conclusions

This article recalled SI models and applications. To provide a thorough narrative of SI models, they were divided into four categories: self-driven particle models, pheromone communication models, empirical research models and leadership decision models. The typical models and applications of each category of models were reviewed. More specifically, PSO and ASO were summarized in self-driven particle models and

pheromone communication models, respectively. Furthermore, the more precise and effective modeling techniques for leadership decisions and empirical research models were described.

The discovery of the mechanism requires rigorous model support, and the model is based on the behavior characteristics of different organisms and the common among the same species of organisms. The existing SI models effectively capture the characteristics of SI and can simulate the collective behavior resembling real biological clusters. These models have demonstrated strong performance at the simulation level. At the level of algorithm, SI algorithms perform well in many applications. However, they also have some limitations, such as issues in stability, parameter tuning, stagnation, the sensitivity to the initial conditions and the dependency of the problem. The issue of stability arises from the intricate dynamics of interactions among individuals within the algorithm, leading to unpredictable outcomes. Parameter tuning complexity stems from the need to find optimal configurations tailored to the unique characteristics of each problem, making it a challenging and problem-specific task. Stagnation poses a significant hurdle, limiting the algorithm's ability to explore and discover global optima, particularly in complex scenarios. Sensitivity to initial conditions further complicates matters, introducing an element of unpredictability in the algorithm's behavior. Additionally, problem dependency highlights the inconsistency in performance across different types of problems, underscoring the need for adaptive algorithms. Successfully addressing these challenges requires not only a nuanced understanding of algorithmic intricacies but also the ability to make domain-specific adjustments enhance their robustness and effectiveness. Issues in stability, parameter tuning and stagnation have been addressed in Ref. 29, while further in-depth study is required for the sensitivity to the initial conditions and the dependency of the problem.

Furthermore, we found that there are still many challenges in the study of SI models and applications need to be broken down in the future, which are summarized as follows.

(1) Model accuracy evaluation system

With the improvement of data analysis technology, data-driven research has become a hotspot in the study of SI. However, the main results of the research are still the simulation verification based on their own designed methods. There is a lack of research on how to quantitatively evaluate the differences between the data driven results and the real collective behavior of biological populations, and a standardized comprehensive evaluation system is a research direction.

(2) Application of heterogeneous populations

The SI has been widely used in intelligent unmanned systems. However, the research on cooperative operation of heterogeneous populations is not efficient enough to cope with the current complex and changeable environment, and the future research direction can be developed to multi-population cooperative perception, information sharing, task allocation, path planning, collaborative decision-making and other aspects.

(3) Unmanned and manned cooperative cluster systems

The research on autonomous unmanned cluster systems has been carried out in many countries, but the research on unmanned and manned cooperative cluster systems is still in the initial stage. It is a feasible research direction to integrate the SI advantage with the system with human participation in decision-making.

CRediT authorship contribution statement

Chao WANG: Writing – original draft, Writing – review & editing. **Shuyuan ZHANG:** Investigation, Writing – review & editing. **Tianhang MA:** Investigation. **Yuetong XIAO:** Investigation. **Michael Zhiqiang CHEN:** Supervision, Writing – review & editing. **Lei WANG:** Funding acquisition, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This study was co-supported by the National Natural Science Foundation of China (No. 61873017) and the Academic Excellence Foundation of Beihang University for PhD Students, China.

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